

**DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
HALF YEARLY EXAMINATION**

CLASS: X
SUBJECT: BIOLOGY [SET A]

FULL MARKS: 80
TIME: 2 HOURS

***Answers to this Paper must be written on the paper provided separately.
You will not be allowed to write during the first 15 minutes.
This time is to be spent in reading the Question Paper.
The time given at the head of this Paper is the time allowed for writing the answers.***

*Attempt all questions from Section I and any four questions from Section II.
The intended marks for questions or parts of questions are given in brackets [].
This question paper consists of 6 printed pages.*

SECTION A (40 Marks)
(Attempt all questions from this Section)

Question 1.

Select the correct answers to the questions from the given options. (Do not copy the question. Write the correct answer only): [15]

- i. A cell has five pairs of chromosomes. After mitotic division, the number of chromosomes in the daughter cells will be:**

 - a. Five,**
 - b. Ten,**
 - c. Twenty,**
 - d. Forty.**
- ii. The specific function of light energy in the process of photosynthesis is to:**

 - a. Reduce Carbon dioxide,**
 - b. Synthesise glucose,**
 - c. Activate chlorophyll,**
 - d. Oxidise Carbon dioxide.**
- iii. The rate of photosynthesis is not affected by:**

 - a. Light intensity,**
 - b. Humidity,**
 - c. Temperature,**
 - d. Carbon dioxide.**
- iv. In cohesion-tension theory, cohesion refers to the:**

 - a. Attraction between water molecules**
 - b. Attraction between the xylem vessel**
 - c. Attraction between water molecules and xylem**
 - d. Attraction between water molecules and phloem.**
- v. Insulin is secreted by**

 - a. Beta cells of Pancreas,**
 - b. Alpha cells of Pancreas,**
 - c. Delta cells of Pancreas,**
 - d. Gamma cells of Pancreas**

- ### Question 2

[5]

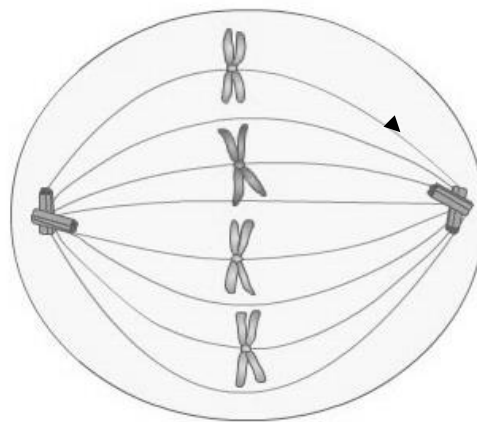
- $A/2$

- e. The part of the chloroplast where the light reaction of photosynthesis takes place.
- (ii) Choose the odd one out and write the correct category to which the others belong: [5]
 Example- Large Intestine, Stomach, Mouth, Brain.
 Odd one- Brain, Category- Organs of Digestive System
- Thyroid, Thyroxine, Pancreas, Adrenal Gland
 - Oxytocin, Ethylene, Cytokinin, Auxin
 - Goitre, Cretinism, Myxoedema, Addison's Disease
 - Needles, Syringe, Dirty dressings, Cobalt 60
 - Glycolysis, Photosynthesis, Photolysis, Phosphorylation
- (iii) Arrange and rewrite the terms in each group in the correct order so as to be in a logical sequence beginning with the term that is underlined. [5]
- Soil water, Xylem, Root hair, Cortex.
 - Refrigerator, Skin cancer, Ozone layer depletion, CFCs.
 - Living cell, Flaccid, Hypertonic, Exosmosis.
 - Transpiration, Ascent of sap, Suction force, Root cell.
 - G₁ phase, Mitosis, S phase, G₂ phase.
- (iv) Match the items given in Column I with the most appropriate ones in Column II and rewrite the correct matching pairs. [5]
- | Column I | Column II |
|------------------------|------------------------------|
| a. Fibrinogen | a. Smog |
| b. Respiratory pigment | b. Water Pollution |
| c. Osmosis | c. Urine |
| d. Urochrome | d. Haemoglobin |
| e. Muslin cloth | e. Blood |
| | f. Semi permeable membrane |
| | g. Freely permeable membrane |
- (v) Mention the exact location of the following: [5]
- Thylakoid
 - Ureter
 - Pituitary Gland
 - Bowman's Capsule
 - Thyroid Gland

SECTION B (40 Marks)
(Attempt any four questions from this section)

Question 3

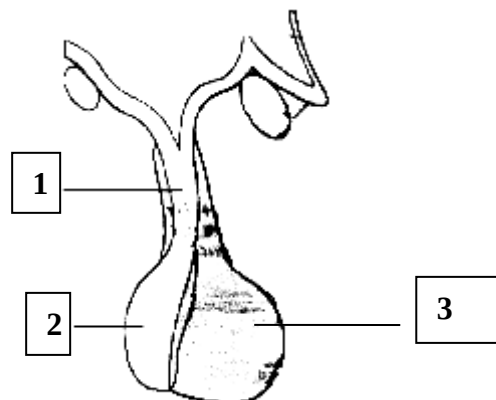
- (i) Define Tropic hormone. [1]
- (ii) Mention the name of the compound formed when carbon monoxide combines with haemoglobin. Why is carbon monoxide harmful for your body? [2]
- (iii) A Tongue roller mother (Tt) and a Non tongue roller father (tt) produces offsprings. Find out the percentage of offsprings who are tongue roller and percentage of offsprings who are non tongue roller. [2]
- (iv) Explain the term Mutation. Name a disease caused in human beings due to gene mutation. [2]
- (v) Observe the diagram given below and answer the following questions: [3]



- a. Identify the stage shown in the above diagram and give a reason to support your answer.
- b. Mention one important feature of the next phase.

Question 4

- (i) Define karyotype. [1]
- (ii) Observe the diagram below and answer the following questions: [2]

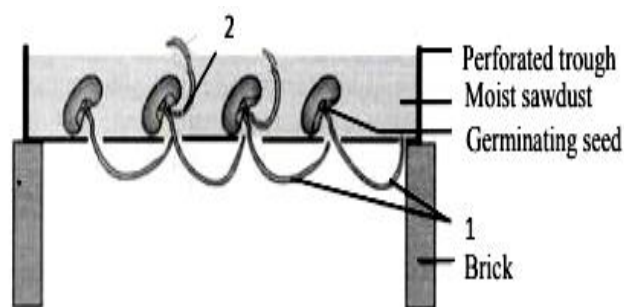


- a. Identify the Endocrine gland shown in the above diagram.
- b. Mention the full form of ACTH.
- (iii) Mention two precautions in the use of a potometer. [2]

- (iv) “Mature Red Blood cells lack nucleus”. Give two reasons to justify this statement. [2]
- (v) Mention one important function of the following: [3]
 - a. Chloroplast
 - b. Cytokinin
 - c. Glucagon

Question 5

- (i) What do you understand by the term Leukopenia? [1]
- (ii) Why do we keep a leaf in boiling water during starch test? What do you understand by the term destarching? [2]
- (iii) Mention any two reasons to state why Mendel had selected Garden Pea for his experiments. [2]
- (iv) Define Parthenocarpy and give one example of a parthenocarpic fruit. [2]
- (v) Study the diagram given below and answer the following questions: [3]



- a. Identify the tropic movements shown by “1” in the diagram.
- b. Which tropism is effective here in case of ‘1’?
- c. Identify the tropic movement observed for ‘2’.

Question 6

- (i) Define pollution. [1]
- (ii) Mention two objectives of Swachh Bharat Abhiyan. [2]
- (iii) Mention two harmful effects of acid rain on monuments and buildings. [2]
- (iv) Differentiate between Mitosis and Meiosis on the basis of [2]
 - a. Number of daughter cells produced.
 - b. Site of occurrence
- (v) Pure Green pod (GG) pea plant is crossed with Pure Yellow pod (gg) pea plant. [3]
 - a. Mention the phenotypic and genotypic ratio of F₂ generation.
 - b. Define Mendel’s Law of Dominance.

Question 7

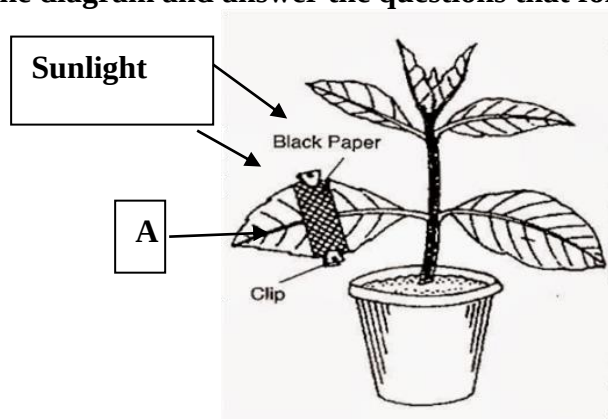
- (i) Mention the scientific name of the plant with which Mendel had conducted his experiments. [1]
- (ii) State briefly about the following abnormal conditions in human body. [2]
 - a. Haematuria
 - b. Glycosuria
- (iii) What do you understand by the term “Destarched Plant”? How is a plant destarched? [2]
- (iv) Define Photosynthesis and write the balanced equation of the process. [2]
- (v) Answer the questions that follow: [3]
 - a. Why are platelets known as thrombocytes?
 - b. Draw a flowchart to show the process of blood clotting when a person is injured.

Question 8

- (i) Mention the full form of ABA. [1]
- (ii) Answer the following questions: [2]



- a. Identify the pollution shown in the diagram given above.
- b. List any two sources of this pollution.
- (iii) Explain the term Chemotropism and give one example of this tropic movement. [2]
- (iv) Study the diagram and answer the questions that follow: [2]



- a. State the aim of the experiment.
- b. Mention what do you observe at the end of the experiment when the leaf labelled 'A' is tested with iodine solution.
- (v) Draw a neat labelled diagram of a chloroplast. [3]